

SYLLABUS

ENSC 429 – Digital Signal Processing

School of Engineering Science
Simon Fraser University
Burnaby, BC, Canada



WELCOME TO ENSC 429

- **Instructor:** Dr. Ivan Bajić
 - Office: ASB 9823
 - Office hours: Thursdays 13:00 – 14:00
- **Teaching Assistants (TAs):**
 - Bardia Azizian
 - Masoud Khairi Atani
 - Soroush Oraki

WELCOME TO ENSC 429

- **Lectures + tutorials:**

- Tuesday 10:30 – 12:20 in SWH 10041
- Thursday 10:30 – 12:20 in C 9002

- **Assignments:**

- ~ Weekly

- **In-class exams:**

- Exam 1 – Thursday, June 25, 2026
- Exam 2 – Tuesday, July 28, 2026

- **Final exam**

- None! 😊

GRADING

- Final grade structure

Assignments	15%
Exam 1	25%
Exam 2	35%
Project	25%
Total	100%

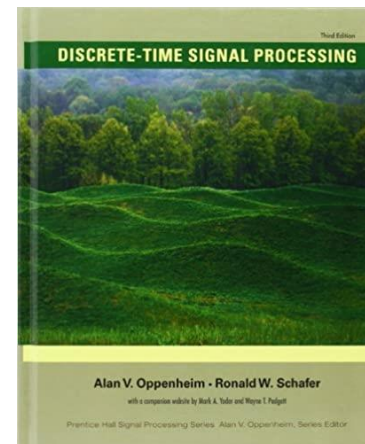
- If you miss Exam 1 due to a **documented** medical reason, the weight will be reassigned to Exam 2, otherwise the exam grade will be 0.
- Participation in Exam 2 is mandatory

COURSE POLICIES

- Due to the large number of students in the course, **neither instructor nor the TAs will be able to respond to e-mailed questions on course material**
 - Please post your questions in the appropriate discussion on Canvas
 - Ask questions during class
 - Come to instructor's office hours (Thursdays 13:00 – 14:00 in ASB 9823)
- **Doctor's note:**
 - The only acceptable doctor's note is
<http://www.sfu.ca/content/dam/sfu/students/pdf/certificate-of-illness.pdf>
 - Someone might call your doctor to verify
- Read about academic (dis)honesty, especially plagiarism, which implies severe consequences
<http://www.sfu.ca/policies/gazette/student.html>

COURSE POLICIES

- Assignment 0 (Plagiarism and Academic Integrity) is already on Canvas
 - complete **by May 19, 2026**
- **Other Assignments:**
 - Mostly problems from the textbook, some mini-projects in MATLAB
 - You can discuss them with your colleagues, but **must be completed individually**
 - Submit electronically (scan if needed) via Canvas. **No paper copies!**
 - **Late submission penalty: -20% per day**
- **Textbook:** A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd Ed., 2009.
- **Software:**
 - MATLAB



COURSE OBJECTIVES

- **To learn the fundamentals of discrete-time linear system analysis**
 - Convolution
 - Transform techniques
 - Time-domain analysis
 - Frequency-domain analysis
 - Stability
 - Sampling
- **To get introduced to discrete-time filter analysis and design**
 - Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters
 - Realization structures
 - Design techniques
 - Industry-standard tools for filter design
 - Applications

ENSC 429 CALENDAR

Unit	Topic	Chapter
1	Introduction	1
2	Review of discrete-time signals	2
3	Review of Fourier Transform	2
4	The z-transform	3
5	Sampling	4
6	Transform analysis of LTI systems	5
7	Discrete-time filter structures	6
8	Discrete-time filter design	7
9	Discrete Fourier Transform	8
10	Fast Fourier Transform	9